

Zadání

Uvažujme dvojkomponentové reálné pole Φ^A , tj. pole nabývající hodnot v dvojdimenzionálním reálném vektorovém prostoru. Komponenty z tohoto prostoru budeme číslovat velkými písmeny $A, B, \dots = 1, 2$.

Na tomto prostoru mějme skalární součin zadaný pomocí metriky H_{AB}

$$(\Phi, \Psi) = H_{AB} \Phi^A \Psi^B$$

Metrika H_{AB} je nede degenerovaná pozitivně definitní s inverzí H^{AB} a budeme ji používat pro zvyšování a snižování indexů.

Dále mějme tzv. komplexní strukturu kompatibilní s tímto součinem - operátor J^A_B splňující

$$J^2 = -I, \quad (J \cdot \Phi, J \cdot \Psi) = (\Phi, \Psi),$$

kde I je jednotkový operátor. V komponentách to znamená

$$J^A_N J^N_B = -I^A_B, \quad H_{MN} J^M_A J^N_B = H_{AB} \implies J^M_A H_{MB} = -H_{AN} J^N_B,$$

což jsou vztahy využitelné pro komutaci matic H_{AB} a J^A_B .

Pole Φ^K se nazývá nabitě skalární pole. Jeho interakce s elektromagnetickým polem je zachycena akcí

$$S_{\text{NP}}[\Phi^N, A_\mu] = -\frac{1}{2} \int_\Omega [H_{AB} D_\mu \Phi^A D_\nu \Phi^B \eta^{\mu\nu} + M^2 H_{AB} \Phi^A \Phi^B] d^4\Omega,$$

4-potenciál A_μ elektromagnetického pole se skrývá v kovariantní derivaci D definované na polním vektorovém prostoru

$$D_\mu \Phi^K = \nabla_\mu \Phi^K - e A_\mu J^K_L \Phi^L.$$

Zde $\eta_{\mu\nu}$ je Minkowského metrika, ∇_μ je derivace v prostoročasu a e je náboj pole. Poznamenejme, že pro kovektor Σ_K s indexem dole je kovariantní derivace dána

$$D_\mu \Sigma_K = \nabla_\mu \Sigma_K + e A_\mu \Sigma_L J^L_K.$$

Uvažujme volbu jednotek $c = 1$.

- i Variací akce podle Φ^N nalezněte pohybovou rovnici pro pole.
- ii Variací akce podle A_μ nalezněte elektrický 4-tok nesený tímto polem

$$\mathcal{J}_{\text{NP}}^\mu = \frac{\delta S_{\text{NP}}}{\delta A_\mu}.$$

- iii) Ověřte, že akce je invariantní vůči kalibrační transformaci

$$\Phi \rightarrow \tilde{\Phi} = \exp(e\alpha J) \cdot \Phi, \quad A_\mu \rightarrow \tilde{A}_\mu = A_\mu + \nabla_\mu \alpha$$

Zde $\exp(e\alpha J)$ je maticová exponenciála. Při důkazu využijte komutační vlastnosti $J^A{}_B$ s H_{AB} .

- iv) Rozštěpením prostoročasu na čas a prostor naleznete Lagrangián pole $L_{\text{NP}}(\Phi^A, \dot{\Phi}^A)$. Zde je výhodné zavést prostorovou kovariantní derivaci $\vec{D}\Phi^K$ a kovariantní časovou derivaci $D_0\Phi^K$

$$\vec{D}\Phi^K = \vec{\nabla}\Phi^K - e\vec{A}J^K{}_L\Phi^L, \quad D_0\Phi^K = \dot{\Phi}^K + e\phi J^K{}_L\Phi^L,$$

kde ϕ a $\vec{A}$ jsou elektromagnetický skalární a vektorový potenciál.

- v) Určete nábojovou hustotu ρ_{NP} a tok náboje $\vec{j}_{\text{NP}}$ asociované s polem

$$\rho_{\text{NP}} = -\frac{\delta L_{\text{NP}}}{\delta \phi}, \quad \vec{j}_{\text{NP}} = \frac{\delta L_{\text{NP}}}{\delta \vec{A}}.$$

- vi) Naleznete hybnost pole

$$P_K = \frac{\delta L_{\text{NP}}}{\delta \dot{\Phi}^K}.$$

- vii) Napište Hamiltonián pole $H_{\text{NP}}(\Phi^A, P_A)$.

Řešení

- i)

Pro variaci akce $S_{\text{NP}}[\Phi^N, A_\mu]$ podle Φ^N platí:

$$\begin{aligned} S_{\text{NP}}[\Phi^N + \varepsilon \delta \Phi^N, A_\mu] &= -\frac{1}{2} \int_{\Omega} [H_{AB} D_\mu (\Phi^A + \varepsilon \delta \Phi^A) D_\nu (\Phi^B + \varepsilon \delta \Phi^B) \eta^{\mu\nu} \\ &\quad + M^2 H_{AB} (\Phi^A + \varepsilon \delta \Phi^A) (\Phi^B + \varepsilon \delta \Phi^B)] d^4\Omega \\ &= -\frac{1}{2} \int_{\Omega} [H_{AB} D_\mu \Phi^A D_\nu \Phi^B \eta^{\mu\nu} + M^2 H_{AB} \Phi^A \Phi^B] d^4\Omega \\ &\quad - \varepsilon \int_{\Omega} [H_{AB} D_\mu \Phi^A D_\nu \delta \Phi^B + M^2 H_{AB} \Phi^A \delta \Phi^B] d^4\Omega + O(\varepsilon^2) \end{aligned}$$

$$\begin{aligned}
 \delta S_{\text{NP}} [\Phi^N, A_\mu] &= - \int_{\Omega} [H_{AB} D_\mu \Phi^A D_\nu \delta \Phi^B + M^2 H_{AB} \Phi^A \delta \Phi^B] d^4 \Omega \\
 &= - \int_{\Omega} [H_{AB} D_\mu \Phi^A (\nabla_\nu \delta \Phi^B - e A_\nu J^B_L \delta \Phi^L) \eta^{\mu\nu} + M^2 H_{AB} \Phi^A \delta \Phi^B] d^4 \Omega \\
 &= - \int_{\Omega} [H_{AB} D_\mu \Phi^A \nabla_\nu \delta \Phi^B \eta^{\mu\nu} - H_{AB} D_\mu \Phi^A e A_\nu J^B_L \delta \Phi^L \eta^{\mu\nu} \\
 &\quad + M^2 H_{AB} \Phi^A \delta \Phi^B] d^4 \Omega \\
 &= - \int_{\Omega} [\nabla_\nu (H_{AB} D_\mu \Phi^A \delta \Phi^B \eta^{\mu\nu}) - H_{AB} \nabla_\nu D_\mu \Phi^A \delta \Phi^B \eta^{\mu\nu} \\
 &\quad - H_{AB} D_\mu \Phi^A e A_\nu J^B_L \delta \Phi^L \eta^{\mu\nu} + M^2 H_{AB} \Phi^A \delta \Phi^B] d^4 \Omega \\
 &= - \int_{\partial \Omega} [H_{AB} D_\mu \Phi^A \eta^{\mu\nu}] \delta \Phi^B d^3 \Sigma_\nu + \int_{\Omega} [H_{AB} \nabla_\nu D_\mu \Phi^A \delta \Phi^B \eta^{\mu\nu} \\
 &\quad + H_{AB} D_\mu \Phi^A e A_\nu J^B_L \delta \Phi^L \eta^{\mu\nu} - M^2 H_{AB} \Phi^A \delta \Phi^B] d^4 \Omega \\
 &= \int_{\Omega} [H_{AB} \nabla_\nu D_\mu \Phi^A \eta^{\mu\nu} + H_{AL} D_\mu \Phi^A e A_\nu J^L_B \eta^{\mu\nu} - M^2 H_{AB} \Phi^A] \delta \Phi^B d^4 \Omega
 \end{aligned}$$

Z čehož získáváme vztah, který můžeme dále upravit:

$$\begin{aligned}
 \frac{\delta S_{\text{NP}}}{\delta \Phi^N} &= H_{AN} \eta^{\mu\nu} \nabla_\nu D_\mu \Phi^A + H_{AL} \eta^{\mu\nu} D_\mu \Phi^A e A_\nu J^L_N - M^2 H_{AN} \Phi^A \\
 &= H_{AN} \eta^{\mu\nu} \nabla_\nu (\nabla_\mu \Phi^A - e A_\mu J^A_L \Phi^L) + H_{AL} \eta^{\mu\nu} (\nabla_\mu \Phi^A - e A_\mu J^A_M \Phi^M) e A_\nu J^L_N \\
 &\quad - M^2 H_{AN} \Phi^A \\
 &= H_{AN} \eta^{\mu\nu} \nabla_\mu \nabla_\nu \Phi^A - H_{AN} \eta^{\mu\nu} (\nabla_\nu A_\mu) e J^A_L \Phi^L - H_{AN} \eta^{\mu\nu} (\nabla_\nu \Phi^L) e A_\mu J^A_L \\
 &\quad + H_{AL} \eta^{\mu\nu} (\nabla_\mu \Phi^A) e A_\nu J^L_N - H_{AL} \eta^{\mu\nu} e^2 A_\mu A_\nu J^A_M J^L_N \Phi^M - M^2 H_{AN} \Phi^A
 \end{aligned}$$

Za použití identit plynoucích ze vztahů pro J a ze symetrie metrik:

$$\begin{aligned}
 H_{AL} \eta^{\mu\nu} (\nabla_\mu \Phi^A) e A_\nu J^L_N &= H_{LA} \eta^{\nu\mu} (\nabla_\nu \Phi^L) e A_\mu J^A_N = -H_{AN} \eta^{\mu\nu} (\nabla_\nu \Phi^L) e A_\mu J^A_L \\
 H_{AL} \eta^{\mu\nu} e^2 A_\mu A_\nu J^A_M J^L_N \Phi^M &= H_{MN} \eta^{\mu\nu} e^2 A_\mu A_\nu \Phi^M = H_{AN} \eta^{\mu\nu} e^2 A_\mu A_\nu \Phi^A
 \end{aligned}$$

získáváme:

$$\begin{aligned}
 \frac{\delta S_{\text{NP}}}{\delta \Phi^N} &= H_{AN} \eta^{\mu\nu} \nabla_\mu \nabla_\nu \Phi^A - H_{AN} \eta^{\mu\nu} (\nabla_\nu A_\mu) e J^A_L \Phi^L - 2 H_{AN} \eta^{\mu\nu} (\nabla_\nu \Phi^L) e A_\mu J^A_L \\
 &\quad - H_{AN} \eta^{\mu\nu} e^2 A_\mu A_\nu \Phi^A - M^2 H_{AN} \Phi^A
 \end{aligned}$$

Pohybovou rovnici následně získáme položením $\frac{\delta S_{\text{NP}}}{\delta \Phi^N}$ rovné nule. Z nede degenerovanosti metriky H_{AN} pak získáme následující tvar pohybové rovnice:

$$\square \Phi^A - e J^A_L \Phi^L \nabla^\mu A_\mu - 2 e A_\mu J^A_L \nabla^\mu \Phi^L - e^2 A^\mu A_\mu \Phi^A - M^2 \Phi^A = 0$$

ii

Pro variaci akce $S_{\text{NP}}[\Phi^N, A_\mu]$ podle A_μ platí:

$$\begin{aligned} S_{\text{NP}}[\Phi^N, A_\mu + \varepsilon \delta A_\mu] &= -\frac{1}{2} \int_{\Omega} \left[H_{AB} \left(\nabla_\mu \Phi^A - e(A_\mu + \varepsilon \delta A_\mu) J^A_M \Phi^M \right) \right. \\ &\quad \cdot \left(\nabla_\nu \Phi^B - e(A_\nu + \varepsilon \delta A_\nu) J^B_L \Phi^L \right) \eta^{\mu\nu} + M^2 H_{AB} \Phi^A \Phi^B \Big] d^4\Omega \\ &= -\frac{1}{2} \int_{\Omega} \left[H_{AB} \left(\nabla_\mu \Phi^A - e A_\mu J^A_M \Phi^M \right) \left(\nabla_\nu \Phi^B - e A_\nu J^B_L \Phi^L \right) \eta^{\mu\nu} \right. \\ &\quad \left. + M^2 H_{AB} \Phi^A \Phi^B \right] d^4\Omega + \varepsilon \int_{\Omega} \left[H_{AB} \nabla_\mu \Phi^A e J^B_L \Phi^L \eta^{\mu\nu} \right. \\ &\quad \left. - H_{AB} e^2 A_\mu J^A_M \Phi^M J^B_L \Phi^L \eta^{\mu\nu} \right] \delta A_\nu d^4\Omega + O(\varepsilon^2), \end{aligned}$$

z čehož přímo použitím identity pro J získáváme vztah pro elektrický 4-tok:

$$\mathcal{J}_{\text{NP}}^\mu = \frac{\delta S_{\text{NP}}}{\delta A_\mu} = e J^B_L \Phi^L \nabla^\mu \Phi_B - e^2 A^\mu \Phi^A \Phi_A$$

iii

Invarianci akce vůči zadané kalibrační transformaci ověříme platností vztahu:

$$S_{\text{NP}}[\Phi^N, A_\mu] = S_{\text{NP}}[\tilde{\Phi}^N, \tilde{A}_\mu]$$

Nejprve odvodíme identitu vhodnou při dokazování plynoucí z vlastností maticové exponenciály a vlastností J :

$$\begin{aligned} H_{AB} \exp(e\alpha J^A_C) &= H_{AB} \exp(e\alpha J^A_L) H^{LM} H_{MC} = \exp(e\alpha H_{AB} J^A_L H^{LM}) H_{MC} \\ &= \exp(-e\alpha J^A_B H_{LA} H^{LM}) H_{MC} = \exp(-e\alpha J^A_B H_{AL} H^{LM}) H_{MC} \\ &= \exp(-e\alpha J^A_B I^M_A) H_{MC} = H_{AC} \exp(-e\alpha J^A_B) \end{aligned}$$

Za použití tohoto vztahu získáváme:

$$\begin{aligned} S_{\text{NP}}[\tilde{\Phi}^N, \tilde{A}_\mu] &= -\frac{1}{2} \int_{\Omega} \left[H_{AB} \left(\nabla_\mu \left(\exp(e\alpha J^A_C) \Phi^C \right) - e(A_\mu + \nabla_\mu \alpha) J^A_D \exp(e\alpha J^D_C) \Phi^C \right) \right. \\ &\quad \cdot \left(\nabla_\nu \left(\exp(e\alpha J^B_L) \Phi^L \right) - e(A_\nu + \nabla_\nu \alpha) J^B_M \exp(e\alpha J^M_L) \Phi^L \right) \eta^{\mu\nu} \\ &\quad \left. + M^2 H_{AB} \exp(e\alpha J^A_C) \Phi^C \exp(e\alpha J^B_D) \Phi^D \right] d^4\Omega \\ &= -\frac{1}{2} \int_{\Omega} \left[H_{AB} \eta^{\mu\nu} \left(\exp(e\alpha J^A_C) \nabla_\mu \Phi^C + \exp(e\alpha J^A_L) e J^L_C \Phi^C \nabla_\mu \alpha - e A_\mu J^A_D \right. \right. \end{aligned}$$

$$\begin{aligned}
& \cdot \exp(e\alpha J^D_K)\Phi^K - eJ^A_D \exp(e\alpha J^D_C)\Phi^C \nabla_\mu \alpha \Big) \Big(\exp(e\alpha J^B_L)\nabla_\nu \Phi^L \\
& + \exp(e\alpha J^B_M)eJ^M_L \Phi^L \nabla_\nu \alpha - eA_\nu J^B_M \exp(e\alpha J^M_L)\Phi^L - eJ^B_M \\
& \cdot \exp(e\alpha J^M_L)\Phi^L \nabla_\nu \alpha \Big) + M^2 H_{AC} \exp(-e\alpha J^A_B) \exp(e\alpha J^B_D)\Phi^C \Phi^D \Big] d^4\Omega \\
& = -\frac{1}{2} \int_\Omega \Big[H_{AB} \Big(\exp(e\alpha J^A_C)\nabla_\mu \Phi^C + eJ^A_D \exp(e\alpha J^D_C)\Phi^C \nabla_\mu \alpha - eA_\mu J^A_D \\
& \cdot \exp(e\alpha J^D_K)\Phi^K - eJ^A_D \exp(e\alpha J^D_C)\Phi^C \nabla_\mu \alpha \Big) \Big(\exp(e\alpha J^B_L)\nabla_\nu \Phi^L \\
& + eJ^B_M \exp(e\alpha J^M_L)\Phi^L \nabla_\nu \alpha - eA_\nu J^B_M \exp(e\alpha J^M_L)\Phi^L - eJ^B_M \\
& \cdot \exp(e\alpha J^M_L)\Phi^L \nabla_\nu \alpha \Big) \eta^{\mu\nu} + M^2 H_{AC} I^A_D \Phi^C \Phi^D \Big] d^4\Omega \\
& = -\frac{1}{2} \int_\Omega \Big[H_{AB} \eta^{\mu\nu} \Big(\exp(e\alpha J^A_C)\nabla_\mu \Phi^C - eA_\mu \exp(e\alpha J^A_C)J^C_K \Phi^K \Big) \\
& \cdot \Big(\exp(e\alpha J^B_L)\nabla_\nu \Phi^L - eA_\nu \exp(e\alpha J^B_L)J^L_M \Phi^M \Big) + M^2 H_{AC} \Phi^C \Phi^A \Big] d^4\Omega \\
& = -\frac{1}{2} \int_\Omega \Big[H_{AB} \exp(e\alpha J^A_C) \exp(e\alpha J^B_L) \Big(\nabla_\mu \Phi^C - eA_\mu J^C_K \Phi^K \Big) \\
& \cdot \Big(\nabla_\nu \Phi^L - eA_\nu J^L_M \Phi^M \Big) \eta^{\mu\nu} + M^2 H_{AB} \Phi^A \Phi^B \Big] d^4\Omega \\
& = -\frac{1}{2} \int_\Omega \Big[H_{AC} \exp(-e\alpha J^A_B) \exp(e\alpha J^B_L) \Big(\nabla_\mu \Phi^C - eA_\mu J^C_K \Phi^K \Big) \\
& \cdot \Big(\nabla_\nu \Phi^L - eA_\nu J^L_M \Phi^M \Big) \eta^{\mu\nu} + M^2 H_{AB} \Phi^A \Phi^B \Big] d^4\Omega \\
& = -\frac{1}{2} \int_\Omega \Big[H_{AC} I^A_L D_\mu \Phi^C D_\nu \Phi^L \eta^{\mu\nu} + M^2 H_{AB} \Phi^A \Phi^B \Big] d^4\Omega = S_{\text{NP}} \left[\Phi^N, A_\mu \right]
\end{aligned}$$

iv

Pro Lagrangian platí vztah:

$$S_{\text{NP}} \left[\Phi^A, A_\mu \right] = \int_{t_Z}^{t_K} L_{\text{NP}} \left(\Phi^A, \dot{\Phi}^A, \phi, \vec{A} \right) dt$$

Lagrangian tak získáme úpravou vztahu pro akci:

$$\begin{aligned}
S_{\text{NP}} \left[\Phi^A, A_\mu \right] &= -\frac{1}{2} \int_\Omega \left[H_{AB} D_\mu \Phi^A D_\nu \Phi^B \eta^{\mu\nu} + M^2 H_{AB} \Phi^A \Phi^B \right] d^4\Omega \\
&= \int_{t_Z}^{t_K} \int_{\Sigma_t} \frac{1}{2} \left[H_{AB} \left(D_0 \Phi^A D_0 \Phi^B - \vec{D} \Phi^A \cdot \vec{D} \Phi^B \right) - M^2 H_{AB} \Phi^A \Phi^B \right] d^3V dt
\end{aligned}$$

Vztah pro Lagrangian můžeme dále upravit:

$$L_{\text{NP}} \left(\Phi^A, \dot{\Phi}^A, \phi, \vec{A} \right) = \int_{\Sigma_t} \frac{1}{2} \left[H_{AB} \left(D_0 \Phi^A D_0 \Phi^B - \vec{D} \Phi^A \cdot \vec{D} \Phi^B \right) - M^2 H_{AB} \Phi^A \Phi^B \right] d^3V$$

$$= \int_{\Sigma_t} \frac{1}{2} \left[H_{AB} \left(\dot{\Phi}^A \dot{\Phi}^B + 2e\phi J^A{}_L \Phi^L \dot{\Phi}^B + e^2 \phi^2 \Phi^A \Phi^B - \vec{\nabla} \Phi^A \cdot \vec{\nabla} \Phi^B \right. \right. \\ \left. \left. + 2e J^A{}_L \Phi^L \vec{A} \cdot \vec{\nabla} \Phi^B - e^2 A^2 \Phi^A \Phi^B \right) - M^2 H_{AB} \Phi^A \Phi^B \right] d^3V$$

□

Z odvozeného tvaru Lagrangiánu získáváme:

$$L_{\text{NP}}(\Phi^A, \dot{\Phi}^A, \phi + \varepsilon \delta\phi, \vec{A}) = \int_{\Sigma_t} \frac{1}{2} \left[H_{AB} \left(\dot{\Phi}^A \dot{\Phi}^B + 2e(\phi + \varepsilon \delta\phi) J^A{}_L \Phi^L \dot{\Phi}^B + e^2 (\phi + \varepsilon \delta\phi)^2 \Phi^A \Phi^B \right. \right. \\ \left. \left. - \vec{\nabla} \Phi^A \cdot \vec{\nabla} \Phi^B + 2e J^A{}_L \Phi^L \vec{A} \cdot \vec{\nabla} \Phi^B - e^2 A^2 \Phi^A \Phi^B \right) - M^2 H_{AB} \Phi^A \Phi^B \right] d^3V \\ = \int_{\Sigma_t} \frac{1}{2} \left[H_{AB} \left(\dot{\Phi}^A \dot{\Phi}^B + 2e\phi J^A{}_L \Phi^L \dot{\Phi}^B + e^2 \phi^2 \Phi^A \Phi^B - \vec{\nabla} \Phi^A \cdot \vec{\nabla} \Phi^B \right. \right. \\ \left. \left. + 2e J^A{}_L \Phi^L \vec{A} \cdot \vec{\nabla} \Phi^B - e^2 A^2 \Phi^A \Phi^B \right) - M^2 H_{AB} \Phi^A \Phi^B \right] d^3V \\ + \varepsilon \int_{\Sigma_t} \left[H_{AB} \left(e J^A{}_L \Phi^L \dot{\Phi}^B + e^2 \phi \Phi^A \Phi^B \right) \right] \delta\phi d^3V + O(\varepsilon^2)$$

Pro nábojovou hustotu ρ_{NP} tak platí:

$$\rho_{\text{NP}} = -\frac{\delta L_{\text{NP}}}{\delta\phi} = -e J^A{}_L \Phi^L \dot{\Phi}_A - e^2 \phi \Phi^A \Phi_A$$

Analogicky pro tok náboje $\vec{j}_{\text{NP}}$:

$$L_{\text{NP}}(\Phi^A, \dot{\Phi}^A, \phi, \vec{A} + \varepsilon \delta\vec{A}) = \int_{\Sigma_t} \frac{1}{2} \left[H_{AB} \left(\dot{\Phi}^A \dot{\Phi}^B + 2e\phi J^A{}_L \Phi^L \dot{\Phi}^B + e^2 \phi^2 \Phi^A \Phi^B - \vec{\nabla} \Phi^A \cdot \vec{\nabla} \Phi^B \right. \right. \\ \left. \left. + 2e J^A{}_L \Phi^L (\vec{A} + \varepsilon \delta\vec{A}) \cdot \vec{\nabla} \Phi^B - e^2 (A + \varepsilon \delta A)^2 \Phi^A \Phi^B \right) - M^2 H_{AB} \Phi^A \Phi^B \right] d^3V \\ = \int_{\Sigma_t} \frac{1}{2} \left[H_{AB} \left(\dot{\Phi}^A \dot{\Phi}^B + 2e\phi J^A{}_L \Phi^L \dot{\Phi}^B + e^2 \phi^2 \Phi^A \Phi^B - \vec{\nabla} \Phi^A \cdot \vec{\nabla} \Phi^B \right. \right. \\ \left. \left. + 2e J^A{}_L \Phi^L \vec{A} \cdot \vec{\nabla} \Phi^B - e^2 A^2 \Phi^A \Phi^B \right) - M^2 H_{AB} \Phi^A \Phi^B \right] d^3V \\ + \varepsilon \int_{\Sigma_t} \left[H_{AB} \left(e J^A{}_L \Phi^L \vec{\nabla} \Phi^B - e^2 \Phi^A \Phi^B \vec{A} \right) \right] \cdot \delta\vec{A} d^3V + O(\varepsilon^2) \\ \vec{j}_{\text{NP}} = \frac{\delta L_{\text{NP}}}{\delta\vec{A}} = e J^A{}_L \Phi^L \vec{\nabla} \Phi_A - e^2 \Phi^A \Phi_A \vec{A}$$

vi

Z tvaru Lagrangiánu opět získáváme:

$$\begin{aligned}
 L_{\text{NP}}(\Phi^A, \dot{\Phi}^A + \varepsilon \delta \dot{\Phi}^A, \phi, \vec{A}) &= \int_{\Sigma_t} \frac{1}{2} \left[H_{AB} \left((\dot{\Phi}^A + \varepsilon \delta \dot{\Phi}^A)(\dot{\Phi}^B + \varepsilon \delta \dot{\Phi}^B) + 2e\phi J^A{}_L \Phi^L (\dot{\Phi}^B + \varepsilon \delta \dot{\Phi}^B) \right. \right. \\
 &\quad \left. \left. + e^2 \phi^2 \Phi^A \Phi^B - \vec{\nabla} \Phi^A \cdot \vec{\nabla} \Phi^B + 2eJ^A{}_L \Phi^L \vec{A} \cdot \vec{\nabla} \Phi^B - e^2 A^2 \Phi^A \Phi^B \right) \right. \\
 &\quad \left. - M^2 H_{AB} \Phi^A \Phi^B \right] d^3V \\
 &= \int_{\Sigma_t} \frac{1}{2} \left[H_{AB} \left(\dot{\Phi}^A \dot{\Phi}^B + 2e\phi J^A{}_L \Phi^L \dot{\Phi}^B + e^2 \phi^2 \Phi^A \Phi^B - \vec{\nabla} \Phi^A \cdot \vec{\nabla} \Phi^B \right. \right. \\
 &\quad \left. \left. + 2eJ^A{}_L \Phi^L \vec{A} \cdot \vec{\nabla} \Phi^B - e^2 A^2 \Phi^A \Phi^B \right) - M^2 H_{AB} \Phi^A \Phi^B \right] d^3V \\
 &\quad + \varepsilon \int_{\Sigma_t} \left[H_{AB} \left(\dot{\Phi}^A + e\phi J^A{}_L \Phi^L \right) \delta \dot{\Phi}^B \right] d^3V + O(\varepsilon^2) \\
 P_K &= \frac{\delta L_{\text{NP}}}{\delta \dot{\Phi}^K} = \dot{\Phi}_K + H_{AK} e\phi J^A{}_L \Phi^L = \dot{\Phi}_K - e\phi J^A{}_K \Phi_A
 \end{aligned}$$

vii

Z předchozího úkolu máme k dispozici vyjádření $\dot{\Phi}^K = \dot{\Phi}^K(\Phi^K, P_K)$. Pro Hamiltonián $H_{\text{NP}}(\Phi^A, P_A)$ pak platí:

$$\begin{aligned}
 H_{\text{NP}}(\Phi^A, P_A) &= \int_{\Sigma_t} \dot{\Phi}^A(\Phi^A, P_A) P_A d^3V - L_{\text{NP}}(\Phi^A, \dot{\Phi}^A(\Phi^A, P_A)) \\
 &= \int_{\Sigma_t} \left[(P^A - e\phi J^A{}_L \Phi^L) P_A \right] d^3V - \int_{\Sigma_t} \frac{1}{2} \left[H_{AB} \left((P^A - e\phi J^A{}_L \Phi^L) (P^B \right. \right. \\
 &\quad \left. \left. - e\phi J^B{}_M \Phi^M) + 2e\phi J^A{}_L \Phi^L (P^B - e\phi J^B{}_M \Phi^M) + e^2 \phi^2 \Phi^A \Phi^B - \vec{\nabla} \Phi^A \right. \right. \\
 &\quad \left. \left. \cdot \vec{\nabla} \Phi^B + 2eJ^A{}_L \Phi^L \vec{A} \cdot \vec{\nabla} \Phi^B - e^2 A^2 \Phi^A \Phi^B \right) - M^2 H_{AB} \Phi^A \Phi^B \right] d^3V \\
 &= \int_{\Sigma_t} \frac{1}{2} \left[H^{AB} P_A P_B - 2e\phi J^A{}_L \Phi^L P_A + H_{AB} \vec{\nabla} \Phi^A \cdot \vec{\nabla} \Phi^B \right. \\
 &\quad \left. - 2H_{AB} eJ^A{}_L \Phi^L \vec{A} \cdot \vec{\nabla} \Phi^B + H_{AB} e^2 A^2 \Phi^A \Phi^B + M^2 H_{AB} \Phi^A \Phi^B \right] d^3V
 \end{aligned}$$

Výsledky

- i $\square \Phi^A - eJ^A{}_L \Phi^L \nabla^\mu A_\mu - 2eA_\mu J^A{}_L \nabla^\mu \Phi^L - e^2 A^\mu A_\mu \Phi^A - M^2 \Phi^A = 0$
- ii $\mathcal{J}_{\text{NP}}^\mu = eJ^B{}_L \Phi^L \nabla^\mu \Phi_B - e^2 A^\mu \Phi^A \Phi_A$
- iii Akce $S_{\text{NP}}[\Phi^N, A_\mu]$ je invariantní vůči zadané kalibrační transformaci.
- iv
$$L_{\text{NP}}(\Phi^A, \dot{\Phi}^A, \phi, \vec{A}) = \int_{\Sigma_t} \frac{1}{2} \left[H_{AB} \left(D_0 \Phi^A D_0 \Phi^B - \vec{D} \Phi^A \cdot \vec{D} \Phi^B \right) - M^2 H_{AB} \Phi^A \Phi^B \right] d^3V$$

$$= \int_{\Sigma_t} \frac{1}{2} \left[H_{AB} \left(\dot{\Phi}^A \dot{\Phi}^B + 2e\phi J^A{}_L \Phi^L \dot{\Phi}^B + e^2 \phi^2 \Phi^A \Phi^B - \vec{\nabla} \Phi^A \cdot \vec{\nabla} \Phi^B + 2eJ^A{}_L \Phi^L \vec{A} \cdot \vec{\nabla} \Phi^B - e^2 A^2 \Phi^A \Phi^B \right) - M^2 H_{AB} \Phi^A \Phi^B \right] d^3V$$
- v $\rho_{\text{NP}} = -eJ^A{}_L \Phi^L \dot{\Phi}_A - e^2 \phi \Phi^A \Phi_A, \vec{j}_{\text{NP}} = eJ^A{}_L \Phi^L \vec{\nabla} \Phi_A - e^2 \Phi^A \Phi_A \vec{A}$
- vi $P_K = \dot{\Phi}_K - e\phi J^A{}_K \Phi_A$
- vii
$$H_{\text{NP}}(\Phi^A, P_A) = \int_{\Sigma_t} \frac{1}{2} \left[H^{AB} P_A P_B - 2e\phi J^A{}_L \Phi^L P_A + H_{AB} \vec{\nabla} \Phi^A \cdot \vec{\nabla} \Phi^B \right. \\ \left. - 2H_{AB} eJ^A{}_L \Phi^L \vec{A} \cdot \vec{\nabla} \Phi^B + H_{AB} e^2 A^2 \Phi^A \Phi^B + M^2 H_{AB} \Phi^A \Phi^B \right] d^3V$$